Compressing dense similarity for dataset splitting with global and local control

Author information pending

Abstract

An evaluation split defines the relationships across which a model must generalize. Controlling total similarity between folds requires a dense pairwise objective whose matrix takes quadratic space to store. We present a splitting framework that controls fold sizes and represents this objective using low-rank coordinates and compact sums within each fold. Across six molecular datasets, the largest observed rank-256 cut error is 0.0070 of the ordinary kernel's off-diagonal similarity mass. Controls with identical groups and fold counts show that optimization in factor space improves separation beyond that obtained by grouping alone. On 32 CPU cores, the plain setting partitions two million fingerprints in a median 4.1 minutes with a 1% capacity corridor. Comparisons with DataSAIL and an ordinary graph partitioner show different trade-offs between separation and cost: reducing overlap across all pairs and excluding close analogues produce different partitions. Kernel powers and capped groups adjust this trade-off, with additional approximation and computational costs. Independent cohorts in a synthetic family model show why the intended transfer task matters: random and family-separated splits support different model-selection decisions. A metadata-only ODAC25 case constructs a count-matched composition holdout covering 37.7 million configurations through 2,050 group representatives. The framework makes similarity-aware splits computationally practical while keeping approximation error, fold-size constraints and the relevance of the split to the intended task explicit.

Keywords: dataset splitting; similarity; Nyström approximation; model evaluation; molecular machine learning.

1 Introduction

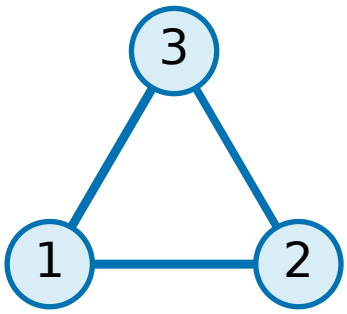
An evaluation split defines the generalization task a model must solve. Predicting another member of a known chemical series differs from predicting members of a new series. Random, scaffold, cluster and temporal splits impose different boundaries between these tasks and can change apparent model performance [Wu et al., 2018, Yang et al., 2019]. Lo-Hi makes the distinction between lead optimization and novel-hit identification explicit [Steshin, 2023]. Temporal studies and SIMPD show why the appropriate boundary depends on the project scenario [Sheridan, 2013, Landrum et al., 2023]. An evaluation procedure therefore needs both a practical way to construct the split and a scientific justification for the generalization task it defines.

Similarity provides a direct objective for split design: reduce similarity between folds while controlling their sizes. The objective includes every entity pair, so storing its matrix requires quadratic memory. DataSAIL addresses the problem through clustering and integer programming [Joeres et al., 2025, 2026]. A nearest-neighbour graph represents selected pairwise relationships. This raises a computational question: can the dense overlap objective be optimized approximately without storing the full matrix?

A low-rank approximation represents pairwise similarities using a smaller set of coordinates. Each entity receives a short vector whose dot product approximates its similarity to another entity. Cross-fold overlap can then be evaluated from the sum of vectors in each fold. The solver uses these compact sums to compare assignments subject to explicit fold-size constraints (Figure 1). This benefit depends on whether the chosen kernel is compressible at an affordable rank. Nyström approximation and its connections to kernel clustering are established [Williams and Seeger, 2000, Dhillon et al., 2004, Gittens and Mahoney, 2016, Wang et al., 2019, Chitta et al., 2011]. Our contribution is a constrained splitting framework, accuracy and feasibility diagnostics, and evidence about the evaluation boundaries it can construct.

Reducing the dense objective does not necessarily exclude close analogues. When similarity is summed over many pairs, a few close analogues in different folds may contribute little to the total. Those analogues may strongly affect prediction. We therefore distinguish global overlap from each test entity's closest training neighbour. Kernel powers emphasize strong similarities, while capped groups encourage nearby entities to remain together. These choices change the optimization problem and its cost. They provide control over an evaluation boundary; their usefulness must be assessed against the intended transfer task.

The experiments examine whether compression preserves the objective and optimization improves separation beyond grouping alone. They also compare separation and cost with established alternatives, measure how fold-size tolerance affects scaling, and test when a split supports a useful evaluation decision. The primary experiments use one pinned implementation. Earlier-release molecular, materials and assay studies are retained as explicitly identified supporting evidence in the Supplementary Information. A separately versioned ODAC25 application tests metadata-only composition holdouts at the original configuration counts (Section 3.5).



Coarsen
Weighted supernodes
and inter-group edges



Sparsify

Select nearby pairs



Compress
Shared coordinates
approximate all pairs

Fold sum $p(c) = \Sigma\ B(i)$ for i in fold c
Cut = ½ [$\|\Sigma\ p(c)\|^2 - \Sigma\ \|p(c)\|^2$]

Figure 1: Representing similarity for split construction. Coarsening replaces entities with weighted supernodes and retains inter-group connections. Sparsification selects individual pairwise edges. Low-rank compression represents similarities through shared coordinates, enabling the fold-sum identity in Equation 2. The illustrations describe representations, not interchangeable feasible sets or identical algorithms.

2 Low-rank similarity splitting

2.1 Objective and compressed representation

Let $S \in \mathbb{R}^{n \times n}$ be a symmetric, nonnegative, positive-semidefinite similarity kernel, and let z_i denote entity i's fold. Define the unordered cross-fold cut and its normalized overlap as

$$\mathcal{C}_S(z) = \sum_{i<j} S_{ij} \mathbf{1}[z_i \neq z_j], \qquad L_S(z) = \frac{\mathcal{C}_S(z)}{\sum_{i<j} S_{ij}}. \tag{1}$$

The denominator is independent of the partition, and diagonal entries do not contribute. We report the ratio only when the denominator is positive. Nonnegative similarity is essential to this interpretation. Positive semidefiniteness alone does not make a signed cosine kernel an overlap measure.

We construct a Nyström factor from r landmarks. If C contains entity–landmark similarities and $W = U\Lambda U^\top$ is the landmark kernel, retaining its numerically positive eigenvalues gives $B = CU_+\Lambda_+^{-1/2}$ and $\widetilde{S} = BB^\top$. For fold sums $p_c = \sum_{i:z_i=c} B_i$, the cut of the approximate kernel is given exactly by

$$\mathcal{C}_{\widetilde{S}}(z) = \frac{1}{2}\left(\left\|\sum_c p_c\right\|^2 - \sum_c \|p_c\|^2\right). \tag{2}$$

Factor storage and fold-sum construction require $O(nr)$ space and work, respectively. This identity makes repeated objective evaluation practical without constructing $\widetilde{S}$.

When fold sizes are equal and fixed, minimizing this cut has the same optima as kernel k-means. Unequal capacities give a different weighting of within-fold similarity, so ordinary kernel k-means guarantees do not transfer directly. We distinguish the approximation of S, the quality of the heuristic optimizer, and the scientific meaning of the selected split. The supplement derives the relevant identities and conditional error bound.

2.2 Capacities, optimization and accuracy

Largest-remainder rounding converts requested proportions into integer targets summing to n. Zero tolerance enforces those targets exactly; positive tolerance permits a declared two-sided corridor. Each seeded restart alternates between calculating fold sums and proposing assignments that satisfy the size constraints. Proposals are accepted only if they do not increase the factor objective. A limited set of candidates is then refined before the final partition is selected. Legal single moves operate within a corridor; swaps preserve exact counts [Fiduccia and Mattheyses, 1982, Kernighan and Lin, 1970]. The standard setting uses four restarts, at most 25 assignment iterations and two refined candidates. The optimizer remains heuristic.

Rank can be fixed or selected using a bounded accuracy budget. The optional joint policy increases the number of sampled pairs when uncertainty prevents a decision. It increases rank when the estimated error is established as too large or the pair-sampling budget is exhausted. It accounts for all checks over a declared rank–sample grid and uses a separate

final audit. The cut criterion measures error for the returned partition, normalized by the kernel's total absolute off-diagonal mass. Establishing this tolerance does not certify a uniform approximation over all partitions or a near-optimal partition. Budget failure is reported explicitly.

2.3 Controlling strong local relationships

For integer $p \geq 1$, we can factorize the elementwise power $S^{\circ p}$. The Schur product theorem preserves positive semidefiniteness. Powering reduces weak similarities relative to strong ones: $0.1^3 = 0.001$, whereas $0.7^3 = 0.343$. If $N_\tau(z)$ counts crossing pairs with similarity at least $\tau > 0$, then $\mathcal{C}_{S^{\circ p}}(z) \geq \tau^p N_\tau(z)$. The powered cut consequently provides a soft penalty on strong crossing pairs. It is distinct from the mean nearest-neighbour evaluation score, and its approximation requirements must be checked separately.

Grouping restricts which assignments are permitted. We process top-32-neighbour edges above a threshold in decreasing similarity, merging components subject to a size cap. A group's factor is the sum of its members' rows; its weight is their count. Group contraction therefore preserves the factor objective exactly. The combined molecular setting uses $p = 3$, threshold 0.4, and a cap of half the smallest target fold, with component growth frozen after a cap-blocked merge.

The size-constraint policy determines what happens when whole groups cannot be assigned within the fold capacities. The default count-first policy may dissolve groups and reports that relaxation. The strict policy preserves every group or reports infeasibility or an unresolved feasibility check. Neither capped grouping nor a powered objective guarantees that every pair above the threshold stays within one fold. All reported separation scores use the original, unpowered kernel.

2.4 Computational scope

The factor avoids storing a quadratic-size matrix, but some computations can still require quadratic work. Input features still require $O(nd)$ space and can dominate process memory. Factor construction, assignment sorting and the number of accepted refinement moves determine runtime. Exact neighbour construction adds quadratic pairwise arithmetic even when temporary memory is blocked. We therefore report scaling separately for the plain and combined settings. Refinement under exact counts can also cost substantially more than refinement within a relaxed corridor. Each timing and memory measurement states which operations it includes.

3 Results

3.1 Compression preserves the ordinary cut, and optimization adds to grouping

We first assess how accurately the factor represents the intended cut objective. Across six full molecular datasets, three seeds and five partitions per factor, the largest observed rank-

256 error is 0.0070 of the ordinary kernel's off-diagonal mass. The corresponding error for the cubed kernel is 0.1367. At rank 512 these values are 0.0027 and 0.0936, respectively (Figure 2). Accuracy for the ordinary kernel therefore cannot be assumed for the powered objective. A coarse powered factor can still produce useful local separation, but that behavior must be measured on the returned partition rather than inferred from ordinary-kernel accuracy.

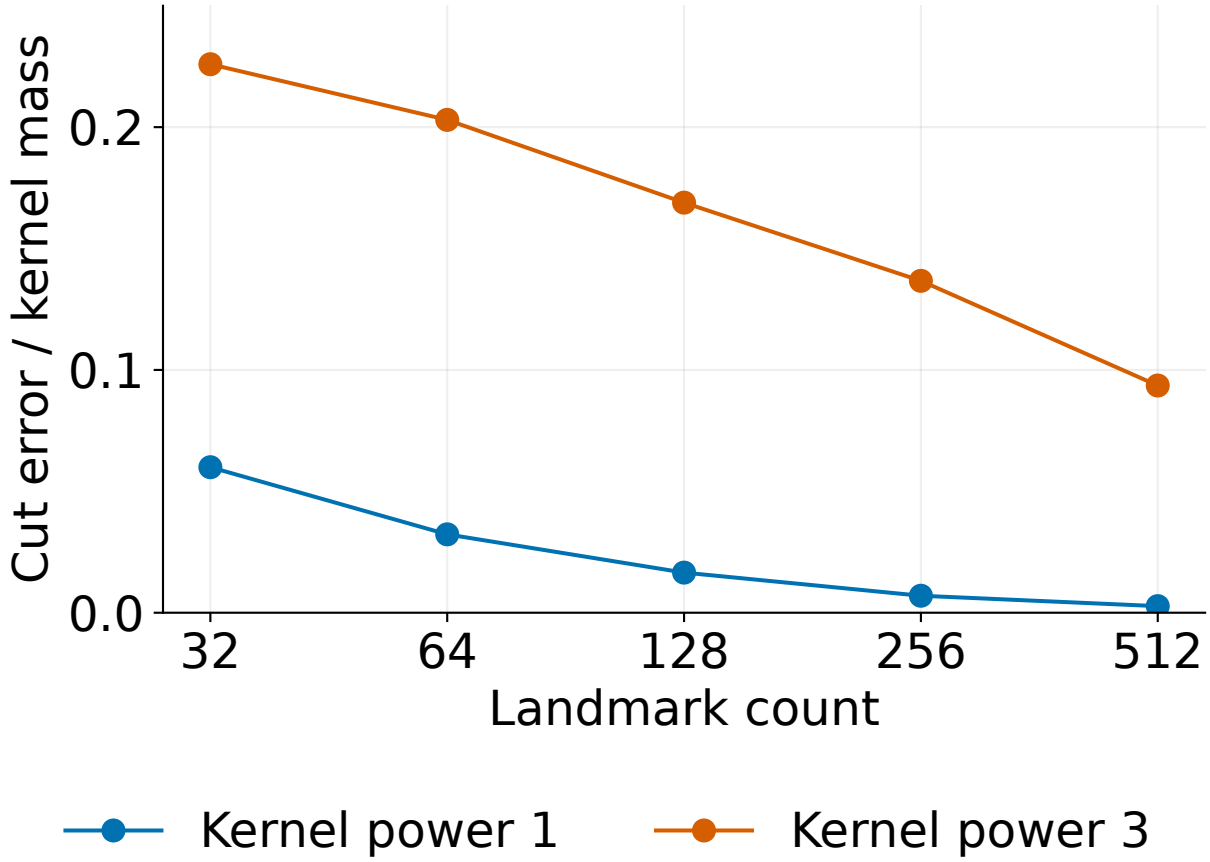


Figure 2: Current-code cut fidelity. Each point is the largest absolute factor-versus-exact cut difference divided by the corresponding kernel's off-diagonal mass, over six full datasets, three factor seeds and five partitions. Each set contains the returned partition and four count-preserving perturbations. The ordinary and cubed kernels have different denominators. These are observed cut errors, not uniform approximation guarantees.

Optimization improves separation beyond that obtained by grouping alone. With the entities, groups and exact fold counts fixed, factor-space optimization improves both global overlap and mean nearest-training similarity on all six datasets, relative to a seeded feasible assignment (Table 1). Every group is preserved in these controls. This control measures the improvement from optimization after groups have been fixed. It does not attribute every advantage of the combined pipeline to compression.

Assignment	Global overlap / random	Mean nearest similarity
Feasible groups	0.959	0.387
Groups + ordinary factor	0.791	0.306
Groups + cubed factor	0.745	0.292

Table 1: Optimization beyond grouping. Every comparison retains the same entities, constructed groups and exact 80/10/10 counts. Values average six dataset means after averaging three seeds. The factor optimizer starts from the feasible control and preserves every group. Both optimized settings improve both measures on all six datasets. Lower values indicate less overlap.

Exhaustive small-instance comparisons separate approximation error from optimization error. On twenty 12-entity instances, we enumerate 2,970 admissible ungrouped assignments and thirty assignments with paired entities as indivisible groups. Even with all twelve landmarks, the heuristic has nonzero optimization gaps on some ungrouped instances: the largest gaps are 0.0132 and 0.0408 of kernel mass for the ordinary and cubed kernels. The grouped full-landmark cases reach their enumerated optima in this small suite. Increasing rank can therefore remove approximation error while leaving a separate optimization error. The complete decomposition and accuracy–cost controls appear in the supplement.

3.2 Dense overlap and analogue exposure give different quality–cost choices

The ordinary factor directly targets similarity summed across all entity pairs. Its mean global score is 0.625, compared with 0.791 for the top-32 graph, after averaging seeds within each dataset. The combined setting reduces nearest-training exposure relative to the plain setting, while the top-32 graph achieves lower mean exposure than the combined setting (0.254 versus 0.282). Changing graph degree also changes the available trade-off (Figure 3). These outcomes support choosing a representation and objective for a specified boundary rather than expecting one method ordering for every separation criterion.

DataSAIL provides a closer comparison for the dense objective and shows why constraints on admissible partitions matter. Sixteen of its twenty-four prespecified configurations return assignments; eight report infeasibility under their cluster and capacity settings. At exactly matched native counts, entity-level low rank has mean global score 0.577, versus 0.666 for DataSAIL. Requiring low rank to preserve the groups passed to DataSAIL's solver changes its mean to 0.668, eliminating that average quality advantage. These summaries weight the four datasets equally after averaging their configurations.

Matching counts alone therefore leaves an important difference in constraints. Matching groups also leaves a distinction between linked cluster similarities and sums over entity-level similarities, as well as different optimizers. The comparison does not establish that compression solves an identical optimization problem better. It demonstrates an economical way to construct splits using entity-level dense overlap, while showing how group indivisibility changes the results. The supplement reports all native outcomes, timings, group-preserving controls and memory profiles.

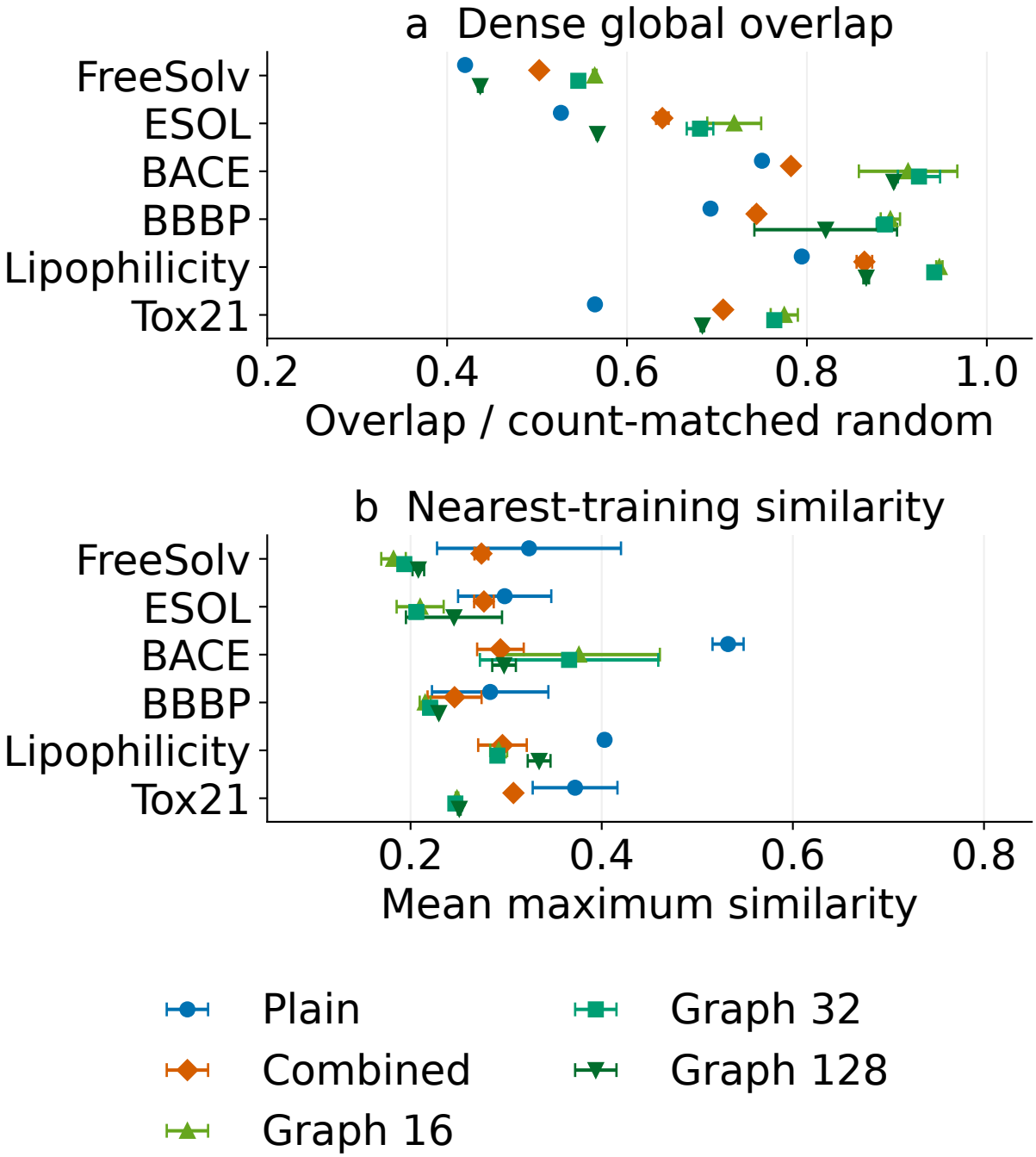


Figure 3: Global and local quality for six full molecular datasets. The primary global score is normalized by the analytic random expectation at the actual counts; the local score is mean test-to-training maximum similarity. Points average three split seeds and bars show their standard deviations. All methods retain the same input entities and use a 1% count corridor. Graph labels identify ordinary pairwise degrees; graph construction and any count repair are included in its runtime. All scores use the original ordinary Tanimoto kernel. Lower values indicate less overlap.

3.3 Exact capacities have a measurable computational cost

At 100,000 ChEMBL fingerprints on two CPU cores, the median plain split-call time is 567.2 seconds with exact counts and 40.1 seconds with a 1% corridor. The inputs, rank and restart settings are otherwise matched. The difference reflects the cost of feasible refinement, including the use of swaps at exact capacities (Figure 4). Linear factor storage therefore does not imply the same runtime at different capacity tolerances.

On lambda13, the current plain implementation partitions two million fingerprints in a median 243.9 seconds using 32 CPU cores, with peak process RSS of 58.8 GiB. All three cases retain all entities and satisfy the declared 1% corridor. This is a separate hardware configuration from the paired two-core comparison; the difference is not interpreted as an algorithmic speedup.

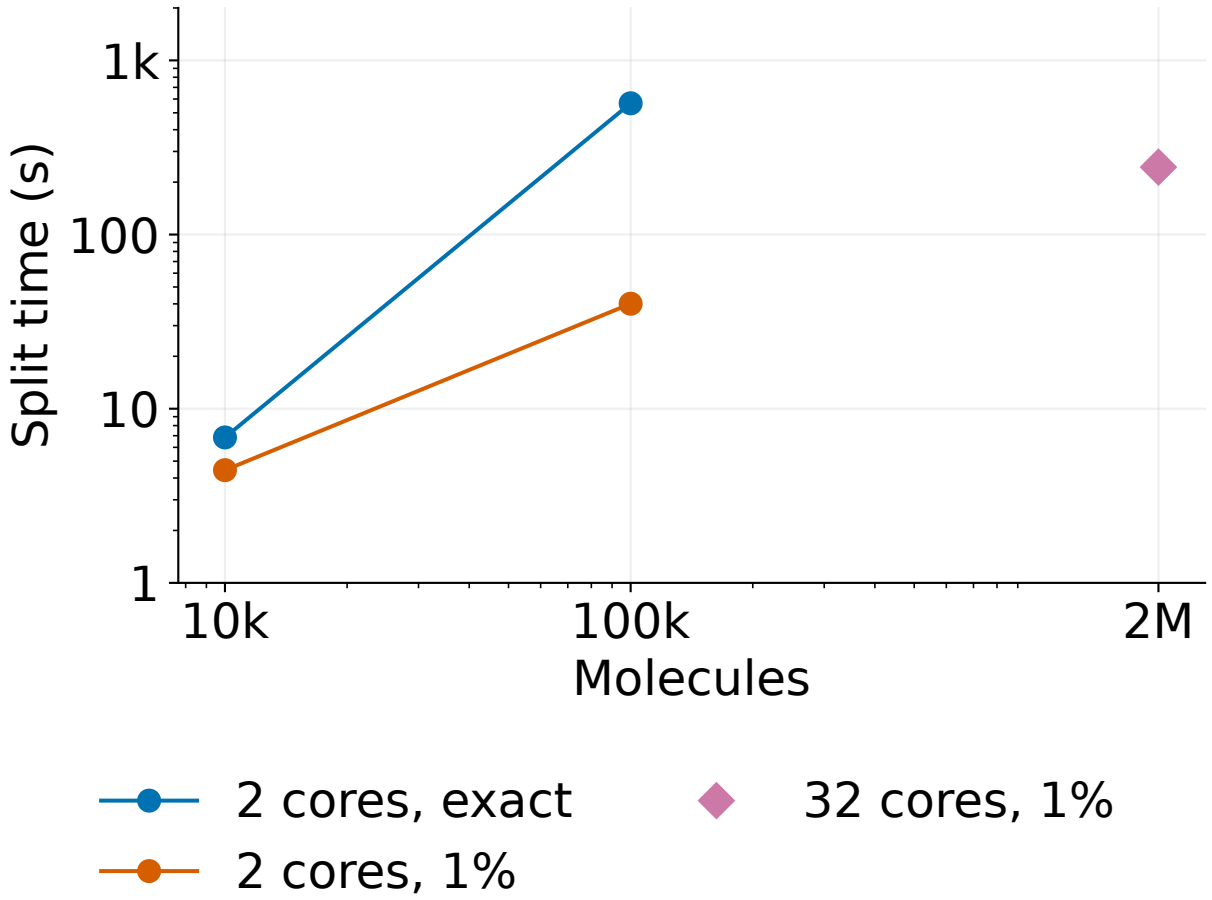


Figure 4: Current-code plain split-call time on nested ChEMBL samples, using rank 256, four restarts and two refined candidates. Two-core cases run on lambda2; the separate 32-core point runs on lambda13. Points are medians of completed cases. Warm-up, feature loading and independent auditing are outside the split clock. The full case inventory, including end-to-end budget failures and the distinct combined-setting range, appears in the supplement. An end-to-end failure is not assigned a split-call runtime.

The large-data protocol evaluates separation and approximation accuracy as well as computational cost. For every completed 100,000- and two-million-entity plain case, the independently sampled returned-cut error interval has an upper endpoint below 0.008 of ordinary kernel mass. Each uniformly sampled test query is compared with every training entity. At two million entities, mean nearest-training similarity is 0.618 across the three sets of sampled queries, showing that low global overlap does not remove close-analogue exposure. Sample sizes and conservative query-sampling intervals accompany the results. The earlier v0.1 H100 measurements remain archival evidence and are not used as timings of the current implementation.

3.4 Split usefulness depends on the transfer task

Independent cohorts allow us to test which transfer task a split reflects. In the synthetic family model, random splitting identifies the better model for known-family interpolation in all twenty generated instances, but identifies the better model for new-family transfer in none. The combined split shows the reverse pattern: agreement in all twenty new-family instances and none of the interpolation instances. Plain splitting lies between these outcomes (Figure 5).

The direction of evaluation error also matters. Random evaluation is optimistic about the family-sensitive model on the new-family cohort, whereas the combined split is pessimistic about that model's interpolation performance. Stronger separation is useful when it matches the intended transfer boundary. Its lower held-out score alone does not make it a more realistic test.

An earlier-release assay study illustrates a related risk in real molecular data. Across ten prespecified external assays, the combined setting gives mean random-forest ROC-AUC 0.522. A label-balance sensitivity intervention at unchanged counts raises this to 0.757. The intervention also changes structural composition and may break groups. It therefore cannot isolate the causal contribution of label proportions. These archival results demonstrate why target composition and absolute predictive scores should accompany similarity measures. They supply no observed future outcomes against which to validate a deployment claim.

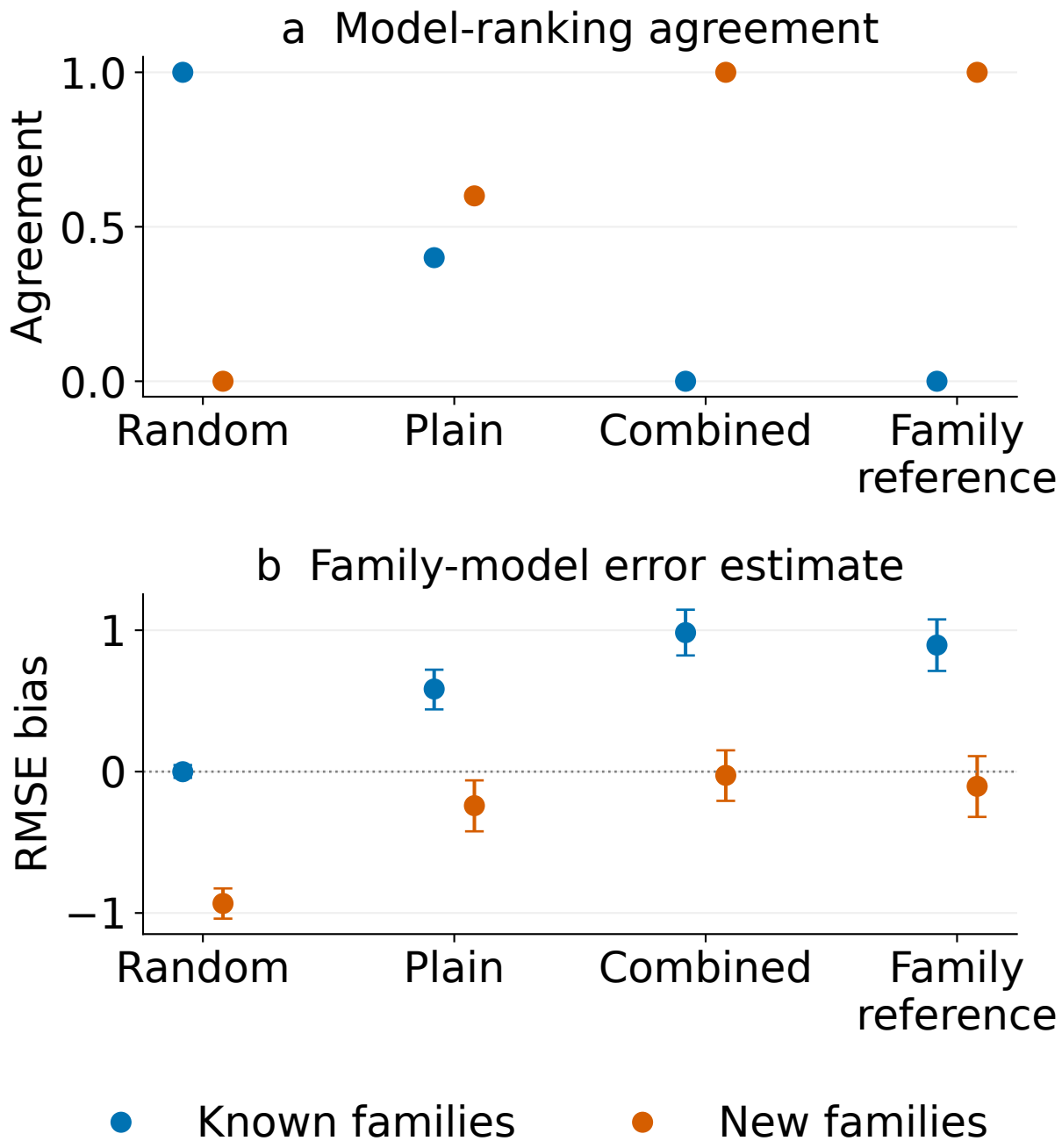


Figure 5: Evaluation boundaries for two transfer tasks in twenty synthetic instances. (a) Agreement of held-out and independent-cohort model rankings. (b) RMSE bias for the family-sensitive model; negative values are optimistic. Bars are descriptive 95% bootstrap intervals over instances. Known-family cohorts condition on families present in training. The family reference uses generator identities, and invariant ridge receives the stable features. This mechanism test supplies no prospective validation.

3.5 A metadata-only composition holdout for ODAC25

The atomistic ODAC25 release provides a separate application of constrained split design [Sriram et al., 2026]. Reading names, formulas and multiplicities from pinned ColabFit Parquet snapshots requires a transfer of 63.4 MB of production payload. After restoring multiplicities, the 37,456,852 retained rows represent 37,711,674 configurations. These totals exactly match the published filtered training and validation counts. Small pilot and documentation requests are additional; coordinates, forces and energy labels are not read. Conservative rules based on family names, shared configuration aliases and identical compositions define 2,050 indivisible components. A versioned weighted-factor adapter then constructs a two-fold holdout at exactly 36,920,036 training and 791,638 validation configurations. This case uses the frozen v0.1 factor implementation and a separate exact-count group assignment, rather than the primary v0.2.0.dev3 point-level solver.

On the recorded elemental-fraction vectors, the count-normalized global overlap falls from 0.969 for the original split to 0.283 after low-rank optimization. Mean nearest-training similarity falls from 0.846 to 0.194–0.196 (Table 2). The same-group feasible initialization already reduces overlap, but further optimization adds substantial separation. A dense-kernel control is competitive, so this small representative problem does not establish a low-rank speed advantage. Thus, the millions of configurations describe how many records receive an assignment; optimization operates on 2,050 representatives.

Assignment	Global ratio ↓	Mean nearest ↓
Original	0.969	0.846
Groups before optimization	0.825–0.873	0.526–0.574
Low-rank optimization	0.283	0.194–0.196
Dense-kernel control	0.287	0.154–0.164

Table 2: ODAC25 composition separation at identical configuration counts. The global ratio is normalized by the random expectation at those counts. Scores use recorded elemental fractions, configuration weights, 200,000 pair draws and 512 validation queries against every distinct training formula. Ranges span three optimizer seeds, which yield only two distinct low-rank assignments. The group control uses the same exact-count feasible initialization; it is not a uniform sampler of feasible partitions.

The resulting boundary is a strong composition-shift test. Phosphorus occurs in 36.8% of validation configurations versus 1.6% of training configurations for seed zero, and two components contribute 40.6% of validation weight. Zero identical- composition overlap follows from the imposed grouping contract. These results demonstrate that the specified holdout can be constructed using a small metadata transfer. Its relevance to prediction and deployment remains untested. The ODAC25 section of the Supplementary Information gives the weighting, aggregation, source versions, individual seeds and numerical checks.

4 Discussion

Low-rank coordinates make dense overlap available as a split-design objective at specified fold capacities. The contribution combines a compact objective, explicit constraints and measurable trade-offs between separation, accuracy and cost. The matched-group controls show that optimization contributes beyond forming groups. The DataSAIL controls also show why feasible sets matter: an advantage observed at matched counts can disappear when indivisible clusters are matched as well. These comparisons support a practical computational framework without establishing universal superiority in partition quality. Compression makes the entity-level objective economical to evaluate, while group construction and size constraints determine the admissible partitions.

Global and local overlap describe different evaluation boundaries. The plain setting favors reducing similarity summed across all entity pairs. Powers and groups can strengthen analogue separation while changing global overlap, approximation accuracy and cost. A sparse graph can provide stronger local separation for a given configuration. Choosing among these outcomes requires specifying which relationships should transfer from training to evaluation. The synthetic independent-cohort experiment demonstrates that both optimistic and pessimistic model-selection errors can result from choosing the wrong boundary.

A practical evaluation should state its intended transfer task, capacities, representation and retention policy before examining final model outcomes. It should then report global overlap, nearest-training exposure and target composition together. The software also permits fold-pair weights when only particular evaluation boundaries matter; the primary benchmarks use the ordinary all-fold objective. Label-informed changes belong to declared benchmark design or sensitivity analysis, with a separate final evaluation.

The ODAC25 application extends this split-design question to an atomistic corpus using a small metadata transfer. It also distinguishes the number of configurations assigned from the number of independent optimization units. The resulting chemical shift, concentration of validation weight in a few components, and competitive dense control limit the interpretation. The case demonstrates a specified composition holdout; computational scaling is established separately by the ChEMBL experiments.

Several limitations remain. The datasets are convenience samples, and conclusions about molecular separation depend on the fingerprint representation. A small approximation error for the returned cut does not establish how close the partition is to the optimum.

Accuracy auditing also has a computational cost. In the three development datasets used for equal-confidence probe comparisons, a fixed 32,768-pair budget was faster than joint adaptive allocation while establishing the same requested tolerance in every seed. Adaptive allocation is therefore an optional budget policy, not an established efficiency improvement. These measurements support retaining fixed probes as the default and auditing the returned partition when changing kernel power or rank.

Powering can require substantially more rank, and strict whole-group capacities can be infeasible. Storage scaling alone does not establish fast construction at every size or tolerance. The synthetic invariant model receives the generative stable features, making it a mechanism test rather than evidence of prospective performance on real projects. Real deployment validation requires independently specified project tasks and genuine dated outcomes; simulated temporal labels cannot supply those outcomes.

5 Methods

5.1 Implementation, data and scoring

Primary molecular experiments use the pinned v0.2.0.dev3 implementation. The ODAC25 case uses a separately frozen weighted adapter, detailed in the ODAC25 section of the Supplementary Information. AI assistance supported code development, analysis scripting and manuscript preparation. Numerical claims are checked against saved observations, source and assignment hashes, independent kernel calculations and targeted regression tests. The ODAC25 case also includes a separate offline replay and numerical audit; its tables are generated from the verified records.

The six full molecular datasets are FreeSolv, ESOL, BACE, BBBP, Lipophilicity and Tox21 [Wu et al., 2018]. Radius-two Morgan fingerprints have 1024 bits, with binary Tanimoto similarity [Rogers and Hahn, 2010, Ralaivola et al., 2005]. Structural optimization does not use response values and retains every input entity for which features were successfully generated. Raw-input, feature, source and assignment hashes identify the exact observations. Earlier-release experiments are separated in the supplement, including the eleven-dataset molecular comparison, two composition-descriptor materials checks, neural-model reuse and external assays.

For actual counts n_c, the random expectation of the global fraction is

$$\mathbb{E}[L_S] = 1 - \frac{\sum_c n_c(n_c - 1)}{n(n-1)}. \tag{3}$$

The primary global score is $L_S/\mathbb{E}[L_S]$. The primary local score is the raw mean NNL $= |T|^{-1} \sum_{i \in T} \max_{j \in R} S_{ij}$, where T is the test fold and R the training fold. Lower values indicate less overlap. The six-dataset comparison evaluates every pair for the global score and searches the full training fold for each test entity's nearest neighbour. Independent NumPy calculations check the saved global and local scores. Threshold ramps and alternative fingerprints are supporting diagnostics; their definitions are kept separate from the primary local score.

5.2 Controls and established comparators

The group mechanism study uses three seeds and exact 80/10/10 counts. For each dataset and seed, a single set of capped groups is fixed. The control is a seeded assignment that satisfies the fold capacities and is constructed without using factor values. Optimization with ordinary and cubed factors starts from this assignment and preserves every group. Exact small-instance references use twenty 12-entity binary datasets, counts 8/2/2, powers one and three, and landmark counts 2, 4, 8 and 12. Enumerating all feasible assignments measures approximation and optimization gaps separately, both without groups and with pairs as indivisible groups.

DataSAIL is the established comparator for reducing dense cross-fold similarity [Joeres et al., 2025, 2026]. An ordinary weighted k-nearest-neighbour graph tests the alternative of selecting pairs rather than compressing their similarities. We use Mt-KaHyPar's Graph API, QUALITY preset and CUT objective [Gottesbüren et al., 2024], with degrees 16, 32 and 128.

Exact top-k construction, safe integer weight encoding, partitioning and any repairs needed to satisfy lower and upper count bounds are included in the split time. Native and repaired assignments are retained separately.

The six-dataset graph and low-rank comparison uses three split seeds, 8:1:1 proportions and tolerance 0.01. DataSAIL 1.3.0 uses C1e with SCIP on FreeSolv, ESOL, BACE and BBBP. Its prespecified grid combines cluster counts 9, $\max(9, \lfloor n/50 \rfloor)$ and $\max(9, \lfloor n/20 \rfloor)$ with solver budgets of 60 and 300 seconds, $\delta = 0.1$ and $\epsilon = 0.01$. For every native assignment, we construct an entity-level low-rank control with identical counts and a second control that also preserves the groups passed to the native solver. Equal groups do not make DataSAIL's linked cluster similarities identical to the entity-level dense objective. We report solver-declared infeasibility as an outcome and do not impute a score.

Lo-Hi, UMAP/Ward, scaffold and Butina comparisons address additional retention or grouping contracts and remain in the explicitly versioned archival studies [Steshin, 2023, Hernández-Hernández and Ballester, 2023, Guo et al., 2025, Bemis and Murcko, 1996, Butina, 1999]. Their retained subsets and realized counts accompany the comparisons. We report these comparisons separately from controls with equal constraints.

5.3 Fidelity, accuracy budgets and computation

We assess full-kernel fidelity on six datasets using three factor seeds, five landmark counts and powers one and three. For each setting, we evaluate the current returned partition and four count-preserving perturbations against an independent full NumPy Tanimoto kernel. Absolute cut error is divided by the off-diagonal mass of the kernel at that power. This yields 900 comparisons, not 900 independent datasets.

Joint accuracy selection uses ranks 32–512, a cut tolerance of 0.05 and probe counts up to 32,768. For R rank stages and Q pair-count stages, selection checks allocate failure probability $0.025/(RQ)$ each. An independent final audit uses failure probability 0.025. A union bound gives at least 95% coverage for selection and final auditing together under the bounded-sampling assumptions. Fixed-size probe grids use the same policy with equal initial and maximum pair counts. Diagnostics record pair counts, allocation decisions and any failure to establish the requested tolerance. Model test outcomes are not used to select rank or audit budgets.

CPU comparisons pin workers to disjoint pairs of cores, limit native thread pools to two threads and disable CUDA visibility. Wall-clock split times include factorization, assignment, refinement and any method-specific graph or clustering work; feature loading, imports and quality scoring are excluded. A small common low-rank warm-up is reported separately. Native memory profiles record fresh-process peak RSS before scoring or matched controls. They include input arrays and framework overhead.

Scaling uses nested samples of packed 2048-bit ChEMBL 37 fingerprints [ChEMBL37], four restarts, two refined candidates, rank 256, and exact or 1% capacity corridors. The plain and combined settings are measured separately. Independent large-data audits use 262,144 uniform pair draws and up to 128 uniformly sampled test queries, each evaluated against the complete training fold. A conservative bounded-mean interval describes query-sampling uncertainty. End-to-end budgets include loading and auditing. Failures to complete within these budgets are reported separately from split-call timings.

5.4 Independent-cohort experiment and uncertainty

A synthetic regression generator supplies fifty observed families with forty entities each. Responses combine a linear function of four stable features, an independent family effect and observation noise. Nonnegative cosine features encode family identity and a weaker continuous signal. Two fixed models represent different inductive assumptions: five-neighbour regression on the full representation and ridge regression supplied with the generative stable features. Ridge regression therefore serves as a privileged invariant reference. Plain and combined splitting use rank 128; the plain cosine kernel uses its exact finite feature map when available.

For twenty prespecified generative seeds, each split retains 1600 training, 200 validation and 200 test entities. Models are fitted on training labels only. We compare held-out RMSE and model rankings with those obtained on two independent cohorts: new observations from families present in the training set, and observations from fifty new families. The family-reference split uses the generator's known family identities and is a mechanism control. Cohort outcomes never enter split construction. This experiment contains no real experiment dates or prospective application evidence.

Summaries average seeds within each dataset before averaging datasets. DataSAIL configuration comparisons likewise average configurations within datasets first. Descriptive bootstrap intervals resample dataset means or independent synthetic instances, using 10,000 draws and seed 31415. Seeds, models and perturbations of the same dataset are not treated as independent projects. All method settings, realized counts, native outcomes and per-case timing and memory scopes are retained.

Data and code availability

Code, manuscript sources, compiled PDFs, study manifests and saved assignments are available in the versioned PALM manuscript release. The primary numerical implementation is pinned to commit `f7e0866`; the delivered source and environment records identify later documentation and manuscript changes and reproduction adapters. Public-input recipes and checksums accompany the records. The earlier v0.1 review release preserves the archival experiments. MoleculeNet, ChEMBL, QMOF, Materials Project and SIMPD inputs retain their original attribution and terms. The ODAC25 addition, frozen adapter, compact evidence and assignment maps are included in `odac25_case` on the ODAC25 revision branch; its dataset sources retain their CC-BY-4.0 attribution. The expanded metadata caches can be regenerated from the pinned column-range requests and are not needed to rebuild the article.

Author contributions

Pending author-supplied CRediT statement.

Funding

Pending author-supplied funding statement.

Competing interests

Pending author-supplied competing-interest statement.

Acknowledgements and use of AI tools

OpenAI Codex and Anthropic Claude assisted with code development and review, analysis and manuscript preparation. The authors are responsible for the scientific claims, references and final submitted manuscript.

References

Zhenqin Wu, Bharath Ramsundar, Evan N. Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S. Pappu, Karl Leswing, and Vijay Pande. MoleculeNet: a benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018. doi: 10.1039/C7SC02664A.

Kevin Yang, Kyle Swanson, Wengong Jin, Connor Coley, Philipp Eiden, Hua Gao, Angel Guzman-Perez, Timothy Hopper, Brian Kelley, Miriam Mathea, Andrew Palmer, Volker Settels, Tommi Jaakkola, Klavs Jensen, and Regina Barzilay. Analyzing Learned Molecular Representations for Property Prediction. *Journal of Chemical Information and Modeling*, 59(8):3370–3388, 2019. doi: 10.1021/acs.jcim.9b00237.

Simon Steshin. Lo-Hi: Practical ML Drug Discovery Benchmark. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/cb82f1f97ad0ca1d92df852a44a3bd73-Abstract-Datasets_and_Benchmarks.html.

Robert P. Sheridan. Time-Split Cross-Validation as a Method for Estimating the Goodness of Prospective Prediction. *Journal of Chemical Information and Modeling*, 53(4):783–790, 2013. doi: 10.1021/ci400084k.

Gregory A. Landrum, Maximilian Beckers, Jessica Lanini, Nadine Schneider, Nikolaus Stiefl, and Sereina Riniker. SIMPD: an algorithm for generating simulated time splits for validating machine learning approaches. *Journal of Cheminformatics*, 15:119, 2023. doi: 10.1186/s13321-023-00787-9.

Roman Joeres, David B. Blumenthal, and Olga V. Kalinina. Data splitting to avoid information leakage with DataSAIL. *Nature Communications*, 16:3337, 2025. doi: 10.1038/s41467-025-58606-8.

Roman Joeres, David B. Blumenthal, and Olga V. Kalinina. Addendum: Data splitting against information leakage with DataSAIL. *Nature Communications*, 17:1597, 2026. doi: 10.1038/s41467-025-67495-w.

Christopher K. I. Williams and Matthias Seeger. Using the Nyström Method to Speed Up Kernel Machines. In *Advances in Neural Information Processing Systems*, volume 13, 2000. URL https://proceedings.neurips.cc/paper/2000/hash/19de10adbaa1b2ee13f77f679fa1483a-Abstract.html.

Inderjit S. Dhillon, Yuqiang Guan, and Brian Kulis. Kernel k-means, Spectral Clustering and Normalized Cuts. In *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 551–556, 2004. doi: 10.1145/1014052.1014118.

Alex Gittens and Michael W. Mahoney. Revisiting the Nyström Method for Improved Large-scale Machine Learning. *Journal of Machine Learning Research*, 17(117):1–65, 2016. URL https://jmlr.org/papers/v17/gittens16a.html.

Shusen Wang, Alex Gittens, and Michael W. Mahoney. Scalable Kernel K-Means Clustering with Nyström Approximation: Relative-Error Bounds. *Journal of Machine Learning Research*, 20(12):1–49, 2019. URL https://jmlr.org/papers/v20/17-517.html.

Radha Chitta, Rong Jin, Timothy C. Havens, and Anil K. Jain. Approximate kernel k-means. In *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 895–903, 2011. doi: 10.1145/2020408.2020558.

C.M. Fiduccia and R.M. Mattheyses. A Linear-Time Heuristic for Improving Network Partitions. In *19th Design Automation Conference*, pages 175–181, 1982. doi: 10.1109/dac.1982.1585498.

B. W. Kernighan and S. Lin. An Efficient Heuristic Procedure for Partitioning Graphs. *Bell System Technical Journal*, 49(2):291–307, 1970. doi: 10.1002/j.1538-7305.1970.tb01770.x.

Anuroop Sriram, Logan M. Brabson, Xiaohan Yu, Sihoon Choi, Kareem Abdelmaqsoud, Elias Moubarak, Pim de Haan, Sindy Löwe, Johann Brehmer, Muhammed Shuaibi, John R. Kitchin, Max Welling, C. Lawrence Zitnick, Zachary Ulissi, Andrew J. Medford, and David S. Sholl. An open dataset of gas adsorption in deformable metal–organic frameworks for direct air capture. *Nature Chemical Engineering*, 2026. doi: 10.1038/s44286-026-00445-5.

David Rogers and Mathew Hahn. Extended-Connectivity Fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Liva Ralaivola, Sanjay J. Swamidass, Hiroto Saigo, and Pierre Baldi. Graph kernels for chemical informatics. *Neural Networks*, 18(8):1093–1110, 2005. doi: 10.1016/j.neunet.2005.07.009.

Lars Gottesbüren, Tobias Heuer, Nikolai Maas, Peter Sanders, and Sebastian Schlag. Scalable High-Quality Hypergraph Partitioning. *ACM Transactions on Algorithms*, 20(1):1–54, 2024. doi: 10.1145/3626527.

Saiveth Hernández-Hernández and Pedro J. Ballester. On the Best Way to Cluster NCI-60 Molecules. *Biomolecules*, 13:498, 2023. doi: 10.3390/biom13030498.

Qianrong Guo, Saiveth Hernandez-Hernandez, and Pedro J. Ballester. UMAP-based clustering split for rigorous evaluation of AI models for virtual screening on cancer cell lines. *Journal of Cheminformatics*, 17(1):94, 2025. doi: 10.1186/s13321-025-01039-8.

Guy W. Bemis and Mark A. Murcko. The Properties of Known Drugs. 1. Molecular Frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996. doi: 10.1021/jm9602928.

Darko Butina. Unsupervised Data Base Clustering Based on Daylight's Fingerprint and Tanimoto Similarity: A Fast and Automated Way To Cluster Small and Large Data Sets. *Journal of Chemical Information and Computer Sciences*, 39(4):747–750, 1999. doi: 10.1021/ci9803381.

ChEMBL37. ChEMBL database release 37, 2026. URL https://doi.org/10.6019/CHEMBL.database.37.